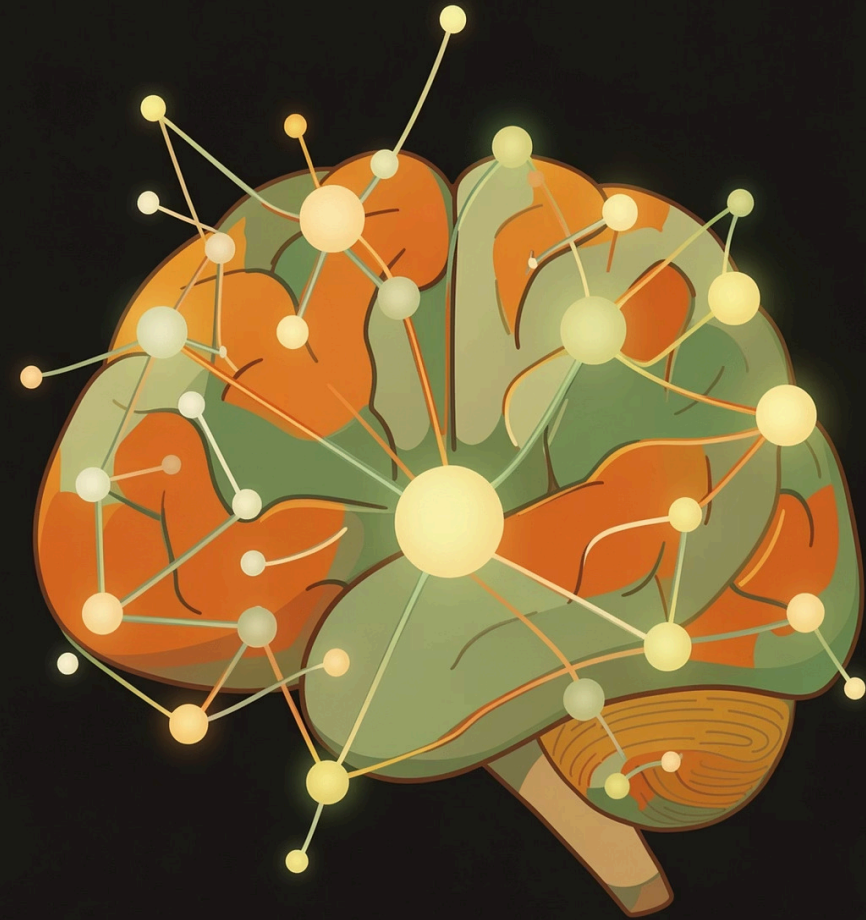


Neuro-Forget

Explainable Alzheimer's Detection with Privacy-Aware Machine Learning

Ahmed Osama



PROBLEM STATEMENT

The Challenge: Early Detection Meets Privacy Risk

🧠 Hard to Detect Early

Alzheimer's progresses silently. By the time symptoms appear, significant brain damage has already occurred.

💰 Expensive & Invasive

MRI and PET scans are costly, resource-intensive, and not scalable for routine screening.

🔒 ML Privacy Risks

Standard ML models memorize training data. Patient records cannot be cleanly deleted — a critical GDPR violation.



OBJECTIVE

What Neuro-Forget Sets Out to Do

An end-to-end biomedical ML system that balances **performance**, **interpretability**, and **patient privacy**.



Predict from Blood

Classify Alzheimer's vs. Control using blood gene expression
— no brain scans required.



Explain Predictions

SHAP-based interpretability reveals which genes drive each decision.



Support Data Deletion

SISA-based machine unlearning removes individual patient influence without full retraining.

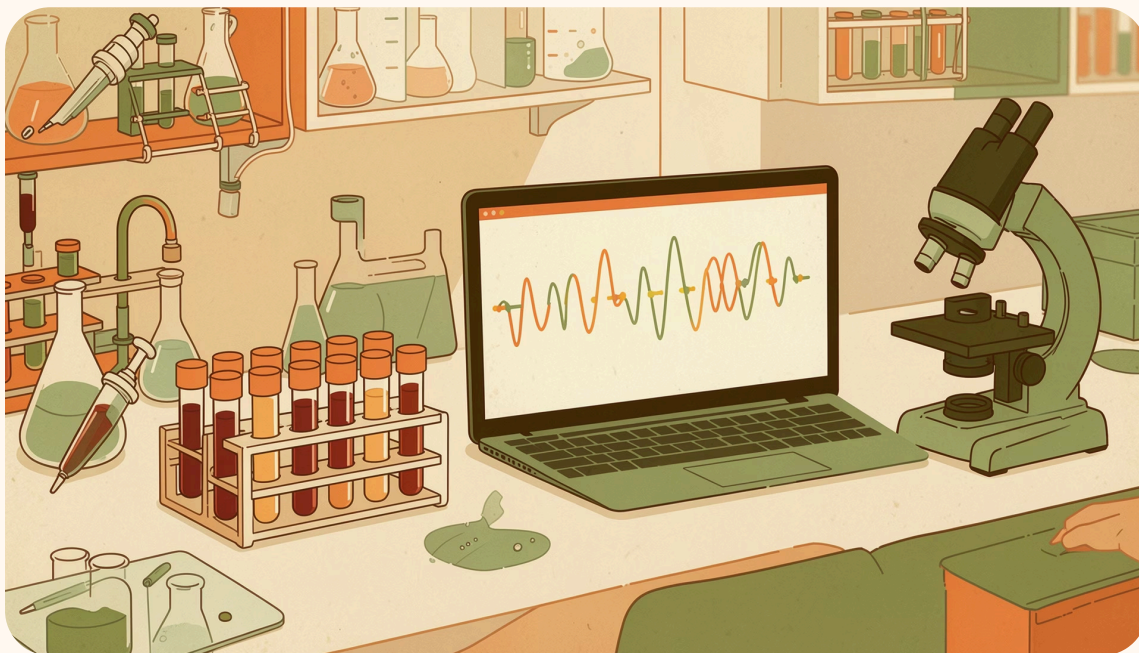


Evaluate Privacy

Membership inference attacks quantify how much the model memorizes training data.

DATASET

GSE63060: Blood Transcriptomics for Alzheimer's



Dataset Profile

Source: GEO public repository — GSE63060

Modality: Blood transcriptomics (RNA expression)

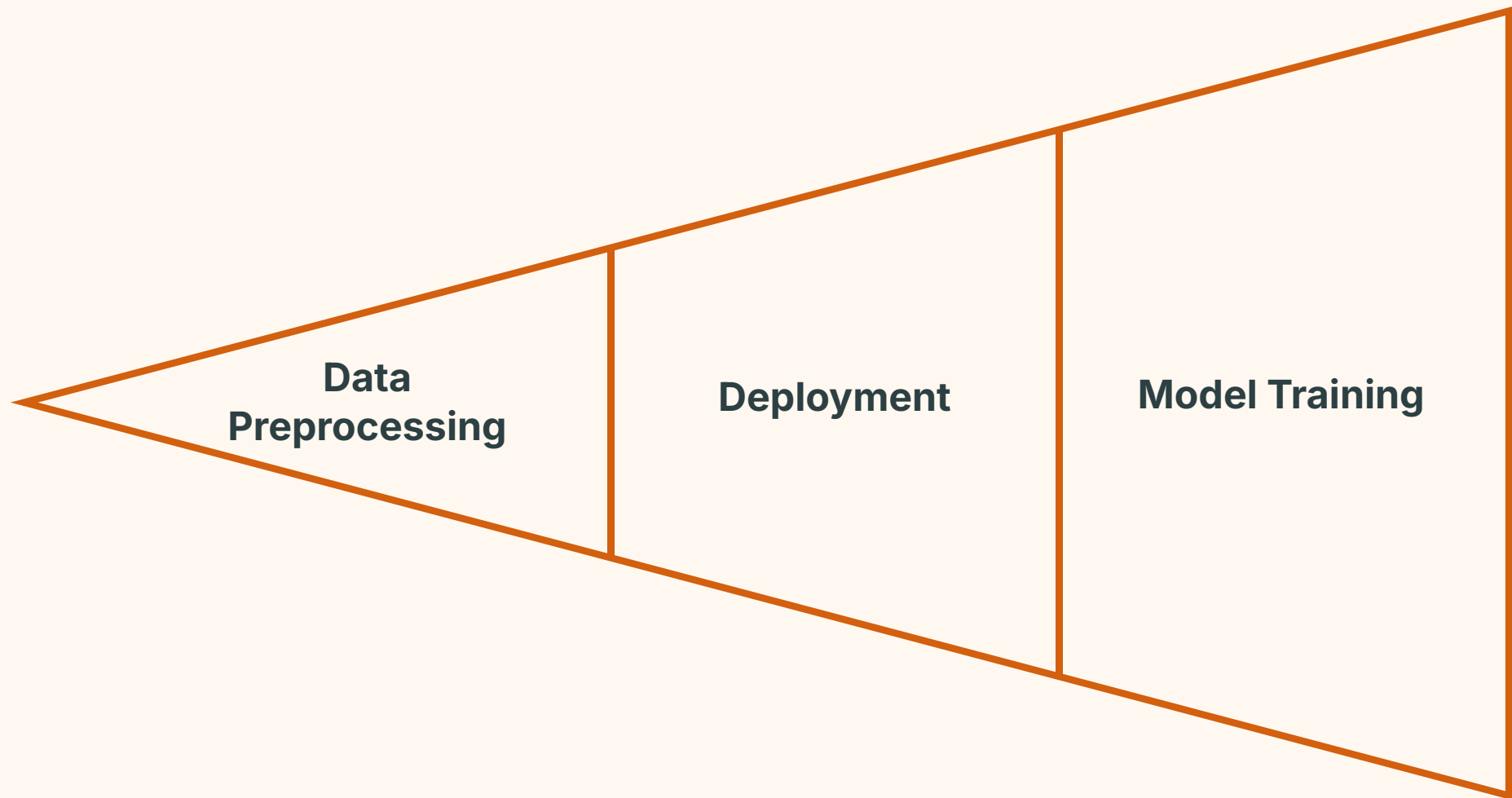
Samples: 329 patients

Features: 136 gene probes

Task: Binary classification — AD vs. Control

- i Blood-based biomarkers offer a non-invasive, scalable alternative to neuroimaging for early-stage screening.

End-to-End Pipeline



Each stage is modular — the system can be audited, updated, or extended independently without breaking downstream components.

Gradient Boosting on Tabular Biological Data

Why XGBoost & LightGBM?

Tabular Strength

Tree-based models dominate structured biomedical data benchmarks.

Non-Linearity

Capture complex gene-gene interactions without manual feature engineering.

Interpretability

Natively compatible with SHAP for post-hoc explanations.

Evaluation Protocol

- Stratified train-test split (class balance preserved)
- 5-fold cross-validation for robust estimates
- Metrics: Accuracy, F1-Score, ROC-AUC

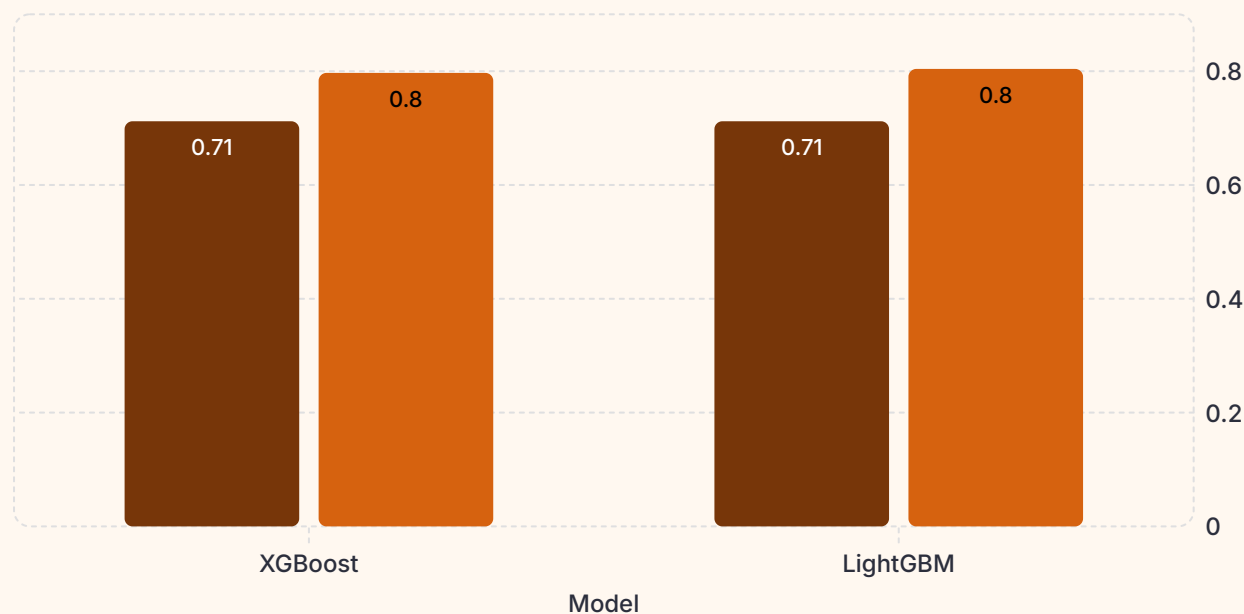
❏ Stratification ensures rare AD cases are represented in every fold.

RESULTS

Strong Discriminative Performance on Biomedical Data

Accuracy

ROC-AUC



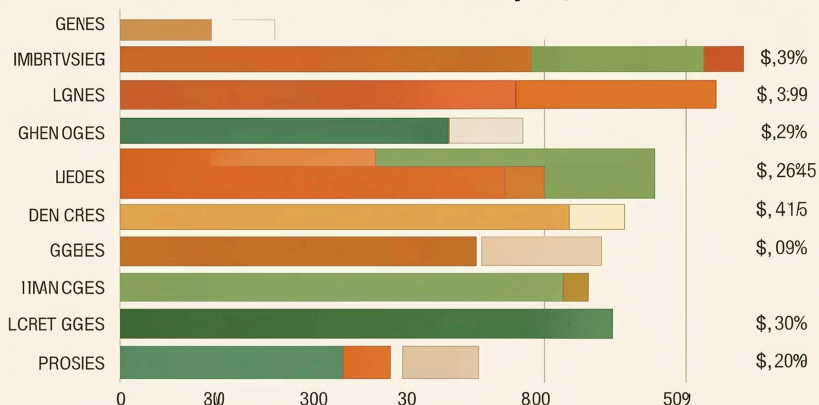
Key Takeaways

Both models achieved **identical accuracy (71.2%)**, with LightGBM edging ahead on ROC-AUC (**0.804 vs. 0.797**).

✅ **ROC-AUC $\approx$ 0.80** is considered strong discriminative power in biomedical classification — especially on a modest 329-sample dataset.

SHAP: Opening the Black Box

SHAP SUMMARCE Tiuvioizs Plot



What SHAP Reveals

SHAP (SHapley Additive exPlanations) assigns each gene a contribution score for every prediction — making model decisions **auditable and clinically interpretable**.

Top Genes Identified

ILMN_1776104 and ILMN_1784286 consistently ranked highest — potential biomarker candidates for further validation.

Not a Black Box

Clinicians can inspect which genes pushed a prediction toward AD or Control for each individual patient.

Machine Unlearning via SISA + Privacy Audit



SISA avoids full retraining by isolating patient data into independent shards — deletion only impacts one sub-model.

SISA Unlearning

Dataset is partitioned into **independent shards**. When a patient requests deletion, only the affected shard is retrained — **fast, scalable, and GDPR-compliant**.

Membership Inference Audit

An adversarial attack tests whether the model **memorizes** training samples. Results showed **measurable confidence leakage** — confirming that privacy-aware ML is not optional in medical AI.

Streamlit UI: Full Interactive Deployment



Model Selection

Toggle between XGBoost and LightGBM



Live Predictions

Input gene expression, get AD probability



SHAP Viz

Visualize gene-level contributions



Patient Deletion

Trigger SISA unlearning on demand



Privacy Audit

Run membership inference tests